University of Basel

File Encryption Project

Project Idea Presentation

Simon Zeugin, Marvin Georges, Nicholas Berger, Frédéric Weyssow

Operating Systems Lecture Spring Semester 2026

March 26, 2026

Outline

Problem Statement

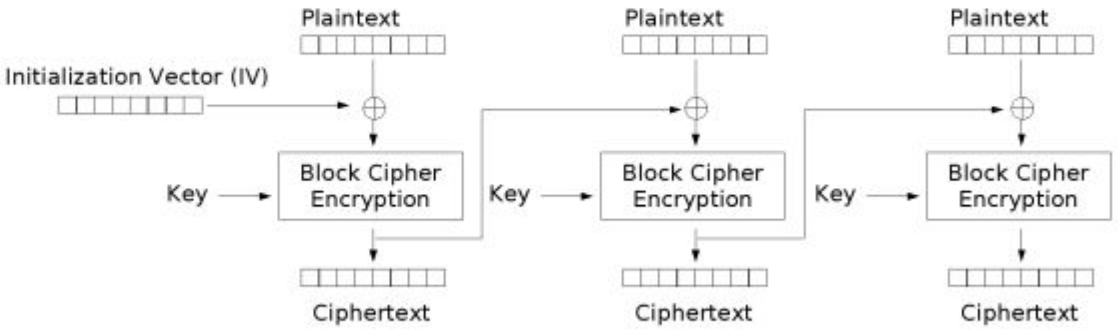
- Ransomware is a huge problem
- Encrypts user data and demands payment for recovery
- Learning by doing

Example: WannaCry

Project description

- Simulate an attack and a way to detect
- Recursively go through all files, starting from home directory
- Encrypting files with multithreading to improve speed
- Key exchange with attacker
- Decrypt files

OpenSSL



Cipher Block Chaining (CBC) mode encryption

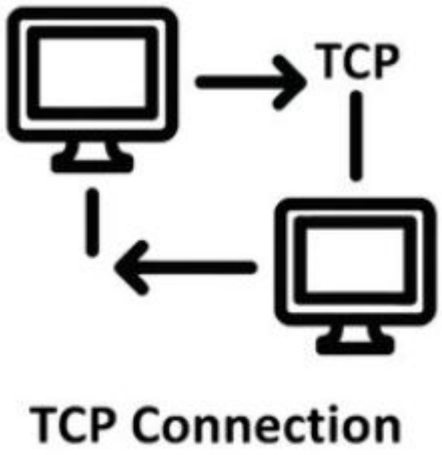
AES-CBC-128

- Software Library
- secure communication over computer networks
- provides several encryption algorithms

Key exchange

- Simple Server/Client TCP connection
- Victim generates encryption key
- Key is sent to attacker via TCP
- Key sent back after victim complies with attacker

- Learn TCP at the OS level with: socket(), bind(), listen(), accept(), connect(), send(), recv(), close()



Monitor Component

Monitoring as a defender:

- Program that monitors the file system
- Detect/kill file encryption process
- Fanotify

Time Plan

- Marvin George, Nicolas Berger
- Frédéric Weyssow, Simon Zeugin
- As a Team